

I chose Beast2 for this lab to estimate a Bayesian phylogeny of primates. I used the mitochondrial DNA alignments and worked with the file provided "primate-mtDNA.nex". I followed the whole BEAST2 tutorial setup which guided me through each step. I did hit some errors after the run finished, so I retraced all my steps and fixed the possible errors. I did have to end up raising the burn-in and increasing a few other values. I visualized the resulting trees and generated the 12 primate taxa including the outgroup.

The BEAST2 analysis it produced a phylogenetic tree showing the evolutionary relationships that the taxa share within the dataset. The tree put the primates together showing humans and chimpanzees closely related as well as the lemur and tarsier lineage more basal related compared to others. Some of the Bayesian analysis might be slightly less resolved depending on the run length, the summary tree still showed an overall pattern for the primate relationships.

This analysis is very interested since it shows the DNA sequence data and how it can reconstruct the evolutionary history between taxas. It is an essential tool for bridging gaps the fossils are currently not able to gap together, it uses genetic information opposed to phenotypically expressed differences to help us understand the divergence of the species. Using BEAST2 was specifically useful because it applies Bayesian methods which help phylogenetic relationships to be estimated when trying to account for uncertainty in the tree. This makes the lab very relevant since it uses computational tools, evolutionary biology, and molecular data for a single method to study how the species are related.